1 Properties of the complex conjugate and the modulus

We will prove the following properties.

(i) $\overline{\overline{z}} = z$

(ii) $\overline{z+w} = \overline{z} + \overline{w}$

(iii) $z + \overline{z} = 2\,\mathrm{Re}(z)$ and $z - \overline{z} = 2i\,\mathrm{Im}(z)$

(iv) $\overline{z \cdot w} = \overline{z} \cdot \overline{w}$

(v) if $w \neq 0$ then $\overline{w} \neq 0$ and

$$\overline{\left(\frac{z}{w}\right)} = \frac{\overline{z}}{\overline{w}}$$

(vi) $z = \overline{z}$ if and only if $z \in \mathbb{R}$

(vii) $z = 0$ if and only if $\overline{z} = 0$

(viii) $|z| = |\overline{z}|$

(ix) $z \cdot \overline{z} = |z|^2$

(x) $|zw| = |z||w|$

(xi) $|\,\mathrm{Re}(z)| \leq |z|$ and $|\,\mathrm{Im}(z)| \leq |z|$

(xii) Triangle inequality: $|z + w| \leq |z| + |w|$

Proof.

(i) Let $z = a + ib$, then $\overline{z} = a - ib = a + i(-b) \Rightarrow \overline{\overline{z}} = a - i(-b) = a + ib = z$

(ii) Let $z = a + ib$ and $w = c + id$, then

$$\overline{z+w} = \overline{a + c + i(b+d)} = a + c - i(b+d) = (a - ib) + (c - id) = \bar{z} + \bar{w}.$$

(iii) Let $z = a + ib$, then

$$z + \bar{z} = a + ib + a - ib = 2a = 2\,\mathrm{Re}(z);\ z - \bar{z} = a + ib - (a - ib) = 2ib = 2i\,\mathrm{Im}(z)$$

(iv) Let $z = a + ib$ and $w = c + id$, then

$$\overline{z \cdot w} = \overline{(a+ib)(c+id)} = \overline{ac - bd + i(ad+bc)} = ac - bd - i(ad+bc) = (a-ib)(c-id) = \bar{z} \cdot \bar{w}$$

(v) Let $w = c + id$, if $w \neq 0$ then either $c \neq 0$ or $d \neq 0$. Since $\bar{w} = c - id$ we thus obtain that $\bar{w} \neq 0$. Next, using property (iv) we have

$$\overline{\left(\frac{z}{w}\right)} \cdot \bar{w} = \overline{\left(\frac{z}{w} \cdot w\right)} = \bar{z}.$$

Dividing both sides by $\bar{w} \neq 0$ gives the result.

(vi) We need to show two implications. First, assume $z = \bar{z}$. Denoting $z = a + ib$ we obtain

$$a + ib = a - ib \;\Rightarrow\; 2ib = 0 \;\Rightarrow\; b = 0 \;\Rightarrow\; z = a \in \mathbb{R}.$$

On the other hand, assume $z = a \in \mathbb{R}$, i.e. $z = a + 0i \Rightarrow \bar{z} = a - 0i = a = z$.

(vii) If $z = a + ib = 0$, then $a = b = 0 \Rightarrow \bar{z} = a - ib = 0$. If $\bar{z} = 0$, then by property (i) $z = \overline{\bar{z}} = 0$ by what we just showed.

(viii) Let $z = a + ib$, then $|z| = \sqrt{a^2 + b^2}$, and $\bar{z} = a - ib \Rightarrow |\bar{z}| = \sqrt{a^2 + (-b)^2} = \sqrt{a^2 + b^2} = |z|$.

(ix) Let $z = a + ib$, then $\bar{z} = a - ib$ so

$$z \cdot \bar{z} = (a + ib)(a - ib) = a^2 + b^2 + iab - iab = a^2 + b^2 = |z|^2.$$

(x) Using (ix) then (iv) and then (ix) again we get

$$|zw|^2 = (zw)\overline{(zw)} = zw\bar{z}\bar{w} = z\bar{z}w\bar{w} = |z|^2|w|^2.$$

Taking the square root on both sides, and using the fact that modulus is always nonnegative, we obtain $|zw| = |z||w|$.

(xi) Let $z = a + ib$, then $|z|^2 = a^2 + b^2$, and $|\operatorname{Re} z|^2 = a^2$. This gives $|\operatorname{Re} z|^2 \leq |z|^2$, which implies by taking the square root and using nonnegativity of modulus, $|\operatorname{Re} z| \leq |z|$. By the property of absolute value ($x \leq |x|\ \forall x \in \mathbb{R}$) we conclude $\operatorname{Re} z \leq |z|$. The second inequality for the imaginary part is shown in the same way.

(xii) For the triangle inequality, we first note that using (iii) for $z\bar{w}$ gives us $z\bar{w} + \bar{z}w = 2\operatorname{Re}(z\bar{w})$ $\forall z, w \in \mathbb{C}$. Note that indeed, using (iv) and then (i) we get $\overline{z\bar{w}} = \bar{z}\bar{\bar{w}} = \bar{z}w$.

Then using (ix), then (ii), then (ix) again, then (xi), and then (viii) and (x) we obtain

$$\begin{aligned} |z + w|^2 &= (z + w)\overline{(z + w)} = (z + w)(\bar{z} + \bar{w}) = z\bar{z} + w\bar{w} + z\bar{w} + \bar{z}w \\ &= |z|^2 + |w|^2 + 2\operatorname{Re}(z\bar{w}) \leq |z|^2 + |w|^2 + 2|z\bar{w}| \\ &= |z|^2 + |w|^2 + 2|z||w| = (|z| + |w|)^2 \end{aligned}$$

Finally, using nonnegativity of the modulus and taking square root of both sides of the inequality completes the proof.

□

Proposition 1.1. For every nonzero $z \in \mathbb{C}$, $\dfrac{z}{|z|}$ is a **unit vector**, i.e. $\left|\dfrac{z}{|z|}\right| = 1$.

Proof. Since $z \neq 0$ then by (vii) $\bar{z} \neq 0$ and thus by (ix) we get $|z| \neq 0$. We can then divide by $|z|$ and using (x) we get

$$|z| = \left|\frac{z}{|z|}|z|\right| = \left|\frac{z}{|z|}\right| \big||z|\big|.$$

We now claim that $\big||z|\big| = |z|$. Indeed, since $|z| \in \mathbb{R}_{\geq 0}$ we have by Remark 3 of Lecture 11 that the modulus of $|z|$ is equal to its absolute value. Since $|z| \geq 0$, by definition of the absolute value we have $\big||z|\big| = |z|$. As a result we get

$$|z| = \left|\frac{z}{|z|}\right| |z|.$$

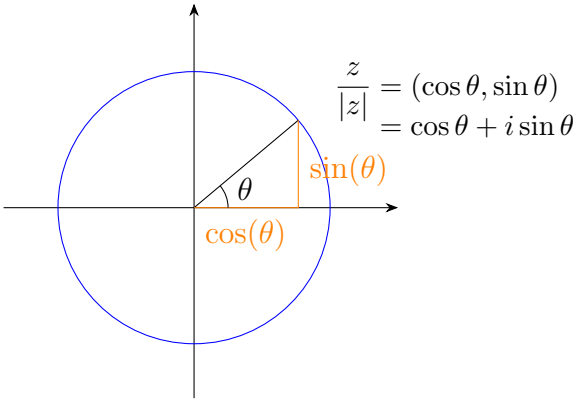
Dividing both sides by $|z| \neq 0$ concludes the proof

$$1 = \left|\frac{z}{|z|}\right|.$$

□

2 Polar form of a complex number

We showed that for any nonzero $z \in \mathbb{C}$, $z/|z|$ has modulus 1. So $z/|z|$ is a point in the complex plane with unit length, i.e., it is a point on the unit circle. In turn, every point on the unit circle has the form $(\cos\theta, \sin\theta)$ where θ is the angle from the positive x-axis to the ray from the origin through the point, measured counterclockwise.



Therefore, identifying points in the plane with complex numbers, we can write

$$\frac{z}{|z|} = (\cos\theta, \sin\theta) = \cos\theta + i\sin\theta$$

for some angle θ. This leads us to the following definition.

Definition 2.1. An *argument* of $z \in \mathbb{C} \setminus \{0\}$ is an angle θ from the positive x-axis to the ray from 0 through z, measured counterclockwise; we write $\arg(z) = \theta$. The *polar form* of z is

$$z = |z|(\cos\theta + i\sin\theta).$$

The polar form describes z in terms of its length $|z|$ and angle θ.

Remark 1. An argument is determined only up to an integer multiple of 2π. When writing $\arg(z) = \theta$, we choose a representative of these angles; a unique choice requires fixing an interval. In these notes we use $[0, 2\pi)$.

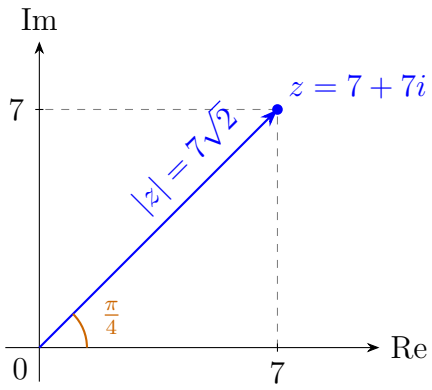
Remark 2. We have $z = |z|(\cos\theta + i\sin\theta) \Rightarrow \operatorname{Re} z = |z|\cos\theta$, $\operatorname{Im} z = |z|\sin\theta$.
Also, if $\operatorname{Re} z \neq 0$, then $\tan\theta = \frac{\operatorname{Im} z}{\operatorname{Re} z}$.
Warning: this does not determine θ, since θ and $\theta + \pi$ have the same tangent. The function $\tan^{-1}$ only returns angles in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, so $\tan^{-1}\left(\frac{\operatorname{Im} z}{\operatorname{Re} z}\right)$ is an argument of z only when z lies in the right half-plane ($\operatorname{Re} z > 0$). Always check which quadrant z lies in.

Remark 3. Later in the course we will talk about the **complex exponential** and represent the circle with radius r as

$$C_r = \{z = re^{i\theta} : 0 \leq \theta \leq 2\pi\}.$$

But we need to talk about Euler's number $e \in \mathbb{R}$ first.

Example 2.1. Let $z = 7 + 7i$.



Then

$$|z| = \sqrt{7^2 + 7^2} = 7\sqrt{2}$$

and

$$\frac{z}{|z|} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i.$$

The point $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$ is the point on the unit circle with angle $45° = \pi/4$. Also note that $\tan\theta = 1$, which alone only gives $\theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$; since z lies in the first quadrant, $\theta = \frac{\pi}{4}$. Thus, the polar form for z is

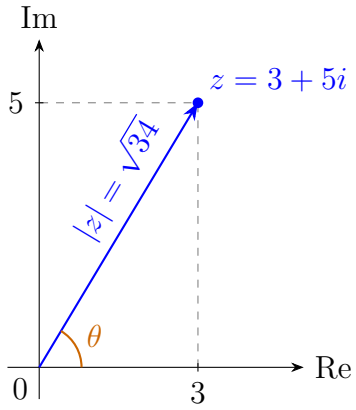
$$z = 7\sqrt{2}\left(\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right).$$

Exercise 2.1. For each of the following complex numbers, find its modulus and an argument $\theta \in [0, 2\pi)$, and write them in polar form:

$$z_1 = \sqrt{3} + i, \qquad z_2 = -1 + i, \qquad z_3 = -\sqrt{3} - i.$$

Sketch each number in the complex plane.

Example 2.2. Let $z = 3 + 5i$.



The length of z is

$$|z| = \sqrt{3^2 + 5^2} = \sqrt{34},$$

and an angle of z is

$$\arg(z) = \tan^{-1}\left(\frac{5}{3}\right) \approx 59^\circ \approx 1.03 \text{ rad}.$$

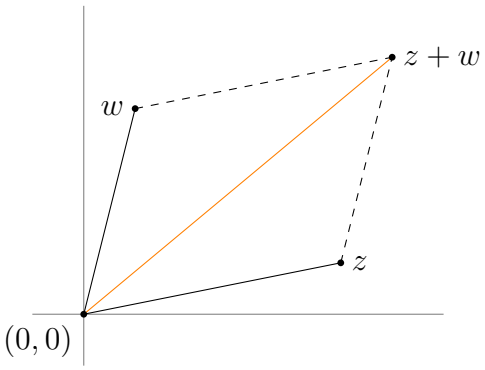
Here $\tan^{-1}$ gives the correct angle because z lies in the first quadrant (see Remark 2). Therefore, the polar form for z is

$$z = \sqrt{34}(\cos\theta + i\sin\theta),$$

where $\theta = \tan^{-1}(5/3)$.

3 Geometry of addition and multiplication in $\mathbb{C}$

The geometry of addition in $\mathbb{C}$ is just the geometry of vector addition.
It is given by the "parallelogram rule":

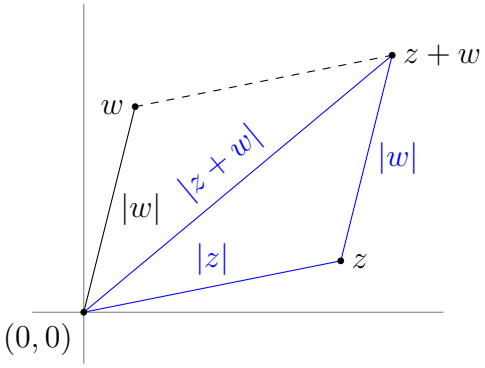


The triangle inequality also has a geometric interpretation.
The points 0, z, and $z + w$ form a triangle with side lengths $|z|$, $|w|$, and $|z + w|$. Indeed, the segment from z to $z + w$ is a translation of the segment from 0 to w, so its length is $|w|$. Since the

length of any side of a triangle is less than or equal to the sum of the lengths of the other two sides, we obtain

$$|z+w| \le |z| + |w|.$$

This explains why it is called the *triangle inequality*.



Question. When does equality hold in the triangle inequality, i.e. for which $z, w \in \mathbb{C}$ do we have $|z+w| = |z| + |w|$? Interpret your answer geometrically.

To see the geometry of multiplication, the secret is to use polar form. Given two nonzero complex numbers $z, w \in \mathbb{C}$, write each in polar form:

$$\begin{aligned} z &= |z|(\cos\theta + i\sin\theta), \\ w &= |w|(\cos\psi + i\sin\psi). \end{aligned}$$

Recall the angle sum formulas for cosine and sine:

$$\begin{aligned} \cos(\theta+\psi) &= \cos\theta\cos\psi - \sin\theta\sin\psi, \\ \sin(\theta+\psi) &= \sin\theta\cos\psi + \cos\theta\sin\psi. \end{aligned}$$

Their product is then

$$\begin{aligned} zw &= |z|(\cos\theta + i\sin\theta)\,|w|(\cos\psi + i\sin\psi) \\ &= |z||w|(\cos\theta + i\sin\theta)(\cos\psi + i\sin\psi) \\ &= |zw|(\cos\theta + i\sin\theta)(\cos\psi + i\sin\psi) \\ &= |zw|\big(\cos\theta\cos\psi - \sin\theta\sin\psi \\ &\qquad + i(\cos\theta\sin\psi + \cos\psi\sin\theta)\big) \\ &= |zw|\big(\cos(\theta+\psi) + i\sin(\theta+\psi)\big). \end{aligned}$$

The last step uses the angle sum formulas for cosine and sine. The last line gives the polar form for zw and reveals the geometry: when multiplying two complex numbers, **angles add and lengths multiply**.

Applying this rule repeatedly (formally, by induction on n) gives *de Moivre's formula*: for every integer $n \ge 1$,

$$\boxed{\big(r(\cos\theta + i\sin\theta)\big)^n = r^n\big(\cos(n\theta) + i\sin(n\theta)\big).}$$

Example 3.1. Let $z = \dfrac{\sqrt{3}}{2} + \dfrac{i}{2}$. Then $|z| = \sqrt{\frac{3}{4} + \frac{1}{4}} = 1$, and since $\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$ and $\sin\frac{\pi}{6} = \frac{1}{2}$, we have $z = 1\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$. By de Moivre's formula we can easily calculate different powers of z

$$\begin{aligned} z^2 &= 1^2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) \\ z^3 &= \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} \\ &\dots \\ z^{13} &= \cos\frac{13\pi}{6} + i\sin\frac{13\pi}{6} = \cos\frac{\pi}{6} + i\sin\frac{\pi}{6} = z. \end{aligned}$$

Question. What is the effect of multiplying a complex number z with i?

Exercise 3.1. Find a formula for i^n for every $n \in \mathbb{N}$ by considering the four possible remainders when n is divided by 4. Plot the first eight terms of the sequence

$$i^0,\ i^1,\ i^2,\ i^3,\ \dots$$

in the complex plane, and interpret the pattern geometrically.

4 Roots of unity

Definition 4.1. Let $n \geq 1$ be an integer. A complex number z is an *nth root of unity* if

$$z^n = 1.$$

To find these roots, we use polar form and de Moivre's formula from Section 3. Suppose $z^n = 1$. Since $|z|^n = |z^n| = 1$, we have $|z| = 1$, so $z = \cos\theta + i\sin\theta$. Then

$$z^n = 1 \quad \Longleftrightarrow \quad \cos(n\theta) = 1 \text{ and } \sin(n\theta) = 0 \quad \Longleftrightarrow \quad n\theta = 2\pi k, k \in \mathbb{Z}.$$

So the nth roots of unity are exactly the numbers $\cos\left(\frac{2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right)$ with $k \in \mathbb{Z}$. Two observations cut this list down:

- Replacing k by $k + n$ adds 2π to the angle, so it gives the same number. Hence every root already appears for some $k \in \{0, 1, \ldots, n-1\}$.
- For $k = 0, 1, \ldots, n-1$ the angles $\frac{2\pi k}{n}$ are distinct and all lie in $[0, 2\pi)$, so they give distinct points on the unit circle.

Thus there are exactly n distinct nth roots of unity, namely

$$\boxed{z_k = \cos\left(\frac{2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right), \qquad k = 0, 1, \ldots, n-1.}$$

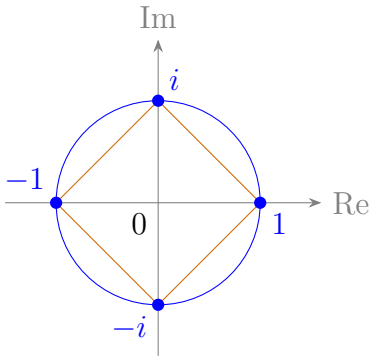
Geometric interpretation

All nth roots of unity lie on the unit circle. Their arguments are

$$0, \ \frac{2\pi}{n}, \ \frac{4\pi}{n}, \ \ldots, \ \frac{2\pi(n-1)}{n},$$

so they are equally spaced, with an angle of $2\pi/n$ between consecutive roots. For $n \geq 3$, they form the vertices of a regular n-gon with one vertex at 1.

Example 4.1. The fourth roots of unity are 1, i, -1, and $-i$. They form the vertices of a square inscribed in the unit circle.



Exercise 4.1. Find all third roots of unity and plot them in the complex plane. Explain why they form the vertices of an equilateral triangle.